

Rohan Shinde

+1-213-234-8020 | rohanshi@usc.edu | [linkedin.com/in/rohanshinde24](https://www.linkedin.com/in/rohanshinde24) | github.com/rohanshinde24 | [Portfolio](#) | Los Angeles, CA

EDUCATION

University of Southern California Los Angeles, CA
Master of Science in Computer Science | GPA: 3.75 Jan 2025 – Dec 2026 (Expected)
Coursework: Analysis of Algorithms, Information Retrieval, Applied NLP, Database Systems, ML for Data Science

Maharashtra Institute of Technology Pune, MH
Bachelor of Technology in Computer Science and Engineering

EXPERIENCE

Amazon May 2026 – Aug 2026
Software Development Engineer Intern Bellevue, WA

- Owned design and deployment of a production Java backend service exposing **four MCP tools** through RPC APIs on **AWS** Lambda, API Gateway, DynamoDB, and S3, with over **95% unit coverage** and integration tests
- Built a **scenario-based evaluation harness** for agent-driven fulfillment investigations with expected tool trajectories and evidence checks, detecting **90% of injected workflow regressions** before deployment
- Reduced average on-call investigation time by **70%**, from 50 to 15 minutes, by integrating the service into an internal troubleshooting platform used by approximately **5,000 users**
- Designed **agent workflow skills** for tool selection and sequencing after working with senior engineers and sister teams to map investigation workflows, edge cases, and data aggregation requirements
- Re-engineered context retrieval across thousands of time-stamped S3 objects per order, replacing paginated listings with bounded DynamoDB date-range queries before fetching relevant evidence

University of Southern California – Advancement Services May 2025 – Feb 2026
Software Engineer Intern Los Angeles, CA

- Built an **event-driven LLM agent system** using Python, Claude, Microsoft Graph, and SQS to turn calendar changes into cited meeting-prep summaries, with idempotent scheduling
- Designed a crash-resumable agent runtime with tool calling, per-task adaptive **model routing** layer, and identity and sensitivity guardrails, achieving zero wrong-person or prompt-injection leaks across **30** live-model evaluations
- Cut p50 latency **10x** and cost per request **80%** under a 200-meeting load test using versioned TTL caching, prompt caching, Batch API pre-processing, and Redis distributed locking across four service replicas

DMI Finance Jul 2024 – Oct 2024
Software Engineer Intern Delhi, India

- Cut synthetic-data generation and labeling turnaround time by **65%** by building a hybrid vLLM and Groq pipeline
- Raised precision from **34%** to **98%** by **fine-tuning** Llama 3.1 with **QLoRA** for 35-class transaction classification
- Re-architected BERT-based NLP pipeline for financial classification, integrating automated training/eval workflows

PROJECTS

- RepoShift** | Python, FastAPI, Docker, Azure OpenAI, GitHub API
- Engineered an **autonomous code-migration system** using **Azure OpenAI**, dependency-graph planning, bounded concurrent workers, and isolated Docker sandboxes with checkpointed, test-guided repair
 - Improved end-to-end migration success from **62% to 84%** over a baseline tool agent and cut median completion time by **41%**, with zero duplicate mutations across **1,000 injected retry and worker-failure scenarios**
- CareRoute** (GitHub) | Python, FastAPI, PostgreSQL, Redis Streams, LangGraph, OpenTelemetry, Docker
- Built a bounded tool-using investigator with deterministic policy validation and a **14-scenario adversarial evaluation suite**, failing closed before unsupported decisions could modify referral state
 - Instrumented agent turns and policy validation with **OpenTelemetry** traces to diagnose model, tool, and execution failures across distributed workflows

TECHNICAL SKILLS

- Languages:** Python, Java, TypeScript, JavaScript, SQL, Go
- Frameworks & APIs:** FastAPI, Spring Boot, React, REST APIs, gRPC
- AI & ML:** LLMs, Tool Calling, LLM Evaluation, Model Routing, RAG, PyTorch, scikit-learn
- Data Systems:** PostgreSQL, Redis, DynamoDB, Elasticsearch, S3, SQS
- Cloud & Infrastructure:** AWS, Docker, Terraform, GitHub Actions, CI/CD, OpenTelemetry, Linux, Git
- Systems:** Distributed Systems, Fault Tolerance, Concurrency, System Design, Data Structures & Algorithms